MAT3032 ANSWERS TO EXERCISES 2.6

Exercises 2.6

1. By Proposition 2.29, if G is simple then it is isomorphic to a subgroup of S_m. Then $|G|$ divides $|S_m| = m!$ so we need the smallest m for which this holds.
   (a) $60 \mid 5!$ so the largest possible subgroup has index 5, hence order 12.
   (b) $360 \mid 6!$ so the largest possible subgroup has index 6, hence order 60.
   (c) $660 \mid 11!$ so so the largest possible subgroup has index 11, hence order 60.
   (d) $2520 \mid 7!$ so the largest possible subgroup has index 7, hence order 360.
   (e) $5616 \mid 13!$ so the largest possible subgroup has index 13, hence order 432.

2. If G has a conjugacy class with two elements, this is the orbit (under the conjugation action) of some element x, so by the orbit-stabiliser theorem $|G| = 2|G_x|$. We know G_x is a subgroup of G so, as it has index 2 in G, it is a proper normal subgroup of G.

   G also has a conjugacy class $\{e\}$ with one element, so $|G| > 2$. Thus G_x is non-trivial. It follows that G is not simple.

3. $20,160 = \dfrac{8!}{2} = |A_8|$. A_8 is a simple group of order 20,160.

   $20,161$ is prime. $C_{20,161}$ is a simple group of order 20,161.

4. (a) Consider the set of all even permutations of $\{1,2,3,4,5,6\}$ which keep 6 fixed. This is a subgroup of A_6 which is isomorphic to A_5, so it has order 60.

   (b) $|A_6| = 360$. If A_6 had a 180-element subgroup, this would have index 2 in A_6 so it would be a normal subgroup of A_6. But we know A_6 is simple, so it has no such subgroup.

5. Suppose $m = 1$, so $|G| = p^k$ where $k > 1$. If G is abelian then any subgroup is normal, in particular one of order p which exists by Cauchy's theorem. If G is non-abelian then by Proposition 2.19 $Z(G)$ is a non-trivial proper subgroup of G, and we know that $Z(G) \lhd G$. Thus G is not simple.

   Now suppose $m > 1$. G has w_p Sylow p-subgroups of order p^k $(\neq |G|)$ where $w_p \equiv 1 \pmod p$ and $w_p \mid m$. As $m < p$ we must have $w_p = 1$ so the p-Sylow is a proper normal subgroup of G.

   Thus there are no simple groups of the following orders, some of which could be shown by other methods: 4, 6, 8, 9, 10, 14, 15, 16, 18, 20, 21, 22, 25, 26, 27, 28, 32, 33, 34, 35, 38, 39, 42, 44, 46, 49, 50, 51, 52, 54, 55, 57, 58.

6. G has a Sylow p-subgroup of order p^k, which has index m in G. By Proposition 2.29, if G is simple then it is isomorphic to a subgroup of S_m. Thus $p^k m$ divides $m!$ so p^k divides $\dfrac{m!}{m} = (m-1)!$ It follows that groups of order 12, 24, 48 cannot be simple, since their orders have the form $2^k.3$ where $2^k \not| \, (3-1)!$

7. G acts on the set of all its p-Sylows by conjugation. The p-Sylows comprise an orbit under this action, whose order is w_p. By the orbit-stabiliser equation, if P is any p-Sylow then $w_p.|G_P| = |G|$ so the stabiliser G_P (which is the normaliser of P in this case), has index w_p in G. (As $w_p > 1$, this subgroup is proper.)

   If G is simple then it is isomorphic to a subgroup of S_{w_p} and so $|G|$ divides $w_p!$

   $36 = 2^2 \times 3^2$. $w_3 = 1$ or 4. As 36 does not divide 4! we must have $w_3 = 1$ so no group of order 36 is simple.

8. If $|G| = p^2q^2$ then $w_p \equiv 1 \pmod{p}$ and $w_p \mid q^2$. The only factors of q^2 are $1, q$ and q^2. As $p > q$ we cannot have $q \equiv 1 \pmod{p}$, so $w_p \neq q$.

   Suppose $w_p = q^2$, so $q^2 \equiv 1 \pmod{p}$, so $q^2 = kp + 1$ for some $k \in \mathbb{Z}$. Then $(q-1)(q+1) = kp$ so $p \mid (q-1)$ or $p \mid (q+1)$. As $p > q$ the former is impossible, and the latter can be true only if $p = q + 1$.

   The only primes for which this holds are 3 and 2, but we are given that p and q are both odd. Thus $w_p \neq q^2$. We conclude that $w_p = 1$ so there is only one p-Sylow. This is normal in G, so G is not simple.

9. Suppose w_p, w_q, w_r are all greater than 1. We have $p > q > r$.

   $w_p \equiv 1 \pmod{p}$ and $w_p \mid qr$. Clearly $w_p \neq q$ and $w_p \neq r$, so $w_p = qr$.

   $w_q \equiv 1 \pmod{q}$ and $w_q \mid pr$. Clearly $w_q \neq r$, so $w_q = p$ or pr, so $w_q \geq p$.

   $w_r \equiv 1 \pmod{r}$ and $w_r \mid pq$. Thus $w_r = p, q$ or pq, so $w_r \geq q$.

   Now the p, q and r-Sylows intersect only in the identity, since their orders are coprime, so the total number of elements that they contain is $1 + w_p(p-1) + w_q(q-1) + w_r(r-1) \geq 1 + qr(p-1) + p(q-1) + q(r-1) = pqr + (p-1)(q-1) > pqr$. But this is greater than $|G|$.

   We have a contradiction, so at least one of w_p, w_q, w_r must be 1. Thus G has a unique, hence normal, p, q or r-Sylow, so G is not simple.

10. (The following are not necessarily the only valid explanations.)

    (a) $30 = 2 \times 3 \times 5$, so no group of order 30 is simple by Question 9.

    (b) $40 = 2^3 \times 5$. $w_5 \equiv 1 \pmod{5}$ and $w_5 \mid 8$, so $w_5 = 1$. Hence there is only one 5-Sylow, which must be normal, so the group is not simple.

    (c) $66 = 2 \times 33$ and 33 is odd, so by Proposition 2.27 no group of order 66 is simple. (Or show that $w_{11} = 1$.)

    (d) $143 = 13 \times 11$ so by Proposition 2.23 every group of order 143 is cyclic. As C_{143} is abelian and not of prime order, it is not simple by Proposition 2.25.

    (e) $225 = 5^2 \times 3^2$, so no group of order 225 is simple by Question 8.

    (f) $132 = 2^2 \times 3 \times 11$. $w_2 = 1, 3, 11$ or 33, $w_3 = 1, 4$ or 22, $w_{11} = 1$ or 12.

    If there is only one 3-Sylow or 11-Sylow, it is normal and so G is not simple.

    Suppose $w_{11} = 12$ and $w_3 > 1$, so there are twelve 11-Sylows and at least four 3-Sylows. These can intersect only in the identity, so together they contain 1 + 120 + 8 = 129 elements. This leaves room for only one 4-element 2-Sylow, which is thus normal. Hence at least one of w_{11}, w_3, w_2 is 1, so G has a normal subgroup and is not simple.

(g) $2430 = 2 \times 3^5 \times 5$. A 3-Sylow, of order 243, has index 10 in G. If G is simple, it is isomorphic to a subgroup of S_{10} so $|G|$ divides 10! But the highest power of 3 that divides 10! is 3^4, so $2430 \not| \ 10!$ Hence G is not simple.